The secrets of success

As invasive species have spread throughout the world, scientists have tried to determine what makes some of these species so successful, and how to control them without accidentally introducing additional ecosystem problems. Despite being hampered by the lack of comparative data on those invasive species that fail to succeed, scientists have developed a number of theories to explain the success of certain species in nonnative environments, including the resource availability hypothesis, the evolution of increased competitive ability hypothesis, and the enemy release hypothesis.

In general, species success depends on a variety of genetic, ecological, and demographic factors. The resource availability hypothesis, first proposed in 1985 by the ecologists Phyllis Coley, John Bryant, and F. Stuart Chapin, argues that an invasive species thrives because it is already well suited to its new environment and can take advantage of any surpluses in resources. The evolution of increased competitive ability hypothesis, published by ecologists Bernd Blossey and Rolf Nötzold in 1995, suggests that invasive plants facing fewer herbivores in their naturalized environment can allocate more resources to reproduction and survival and so outcompete the native species. The enemy release hypothesis, set out by ecologists Ryan M. Keane and Michael J. Crawley in their 2002 article "Exotic Plant Invasions and the Enemy Release Hypothesis," argues that the invasive species has fewer enemies in its naturalized environments, and so can spread farther. The reality is that the success of invasive species is likely due to many mechanisms working together.

"We are seeing one of the great historical convulsions in the world's fauna and flora."

Charles Elton